

Detecting challenge from physiological signals: A primary study with a typical game scenario

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ABSTRACT

Challenge is the core element of digital games. Game challenge with an appropriate type and level that matches with players' skill, experience and motivation would lead players to achieve the optimal player experience. With a wider spectrum of challenge types such as physical, cognitive, and emotional challenge provided by modern digital games, a questionnaire tool of CORGIS has recently been developed to evaluate the whole range of challenge experiences subjectively. However, such challenge experiences still lack measures to evaluate them objectively "in real time". To explore the possibility to detect different challenge types based on physiological signals, we conducted an experiment where 12 players' physiological signals (EDA, ECG, EMG, RSP and TEM) of overcoming different types of game challenges were recorded. With 80 extracted physiological features, two methods (ANOVA-based and Regression-based) were adopted to select challenge-related physiological features. Results of logistic regression models showed that both methods obtained detection accuracy over 60%, which suggest potential for further development of a real-time challenge measurement instrument.

CCS CONCEPTS

• **Human-centered computing** → **User studies; User models.**

KEYWORDS

challenge, digital games, physiological signal analysis, player experience

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CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04.

<https://doi.org/10.1145/3491101.3519806>

ACM Reference Format:

Xiaolan Peng, Chenyu Meng, Xurong Xie, Jin Huang, Hui Chen, and Hongan Wang. 2022. Detecting challenge from physiological signals: A primary study with a typical game scenario. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3491101.3519806>

1 INTRODUCTION

Challenge is the core element of digital games [1]. Different types of challenges require players different abilities to overcome, which further lead to different player experiences (PX) [14]. With the development of game genres, modern digital games are providing not only physical and cognitive types of challenge, but also a new kind of emotional challenge which confronts players with compelling narrative or stories and results in more diverse and eudaimonic PX [4, 10, 11, 14, 24]. One fact is that these challenges rarely appear exclusively in a game. More often, they crisscross in a long game scenario and players need to overcome them one by one. With challenge becoming more complex in digital games, it is important to comprehensively and dynamically evaluate different types of challenges, which would help to understand how different challenges are perceived by players, and further balance challenge types and levels to match with players' skill, experience, and motivation.

Recently, to evaluate the whole range of challenge experiences, a systematic questionnaire of CORGIS has been developed. CORGIS provides a subjective method to quantify how players perceive four types of challenges in video games: cognitive, performative, emotional, and decision making challenge [14]. Before CORGIS, a questionnaire called VGDS has been developed to measure the experience of demand in video games. Similarly, VGDS argued that video games require players' four kinds of faculties: physical, cognitive, social and emotional [5]. However, both CORGIS and VGDS are post-test evaluation methods, which can not evaluate dynamic changes on how players perceive challenge during the game. This makes us only get a global challenge distribution of the

whole game. Many practical needs of game design, such as game process understanding and dynamic challenge adjustment, are not met by CORGIS or VGDS.

Compared to post-test questionnaire, physiological sensing has irreplaceable advantages in real-time assessing of user status. Such methods has long been leveraged to analyze or detect real-time PX, such as the state of flow [2, 3, 19] and multiple game-induced emotions [6, 20, 25]. In our work, we assume that the process in which players make different efforts to overcome different challenges may also lead to different challenge-related physiological responses. Our ultimate goal is to create a real-time challenge measurement tool using physiological signals. This paper reports on a primary study that probe into easily accessible physiological signals that could help distinguish the different types of game challenges. We first choose a Fallout 4 game scenario that provides typical kinds of challenges, and then conducted an experiment in which 12 participants played the game and their physiological signals including electrocardiography (ECG), electrodermal activity (EDA), respiratory activity (RSP), electromyography (EMG) and skin temperature (TEM) were recorded. After extracting 80 physiological features, we adopted two methods to select challenge-related features and compared their regression performance through 10-fold and 10-participant-fold cross validation.

2 RELATED WORK

Challenge often refers to tasks that players need to accomplish in games [1, 26]. Overcoming different types of challenges require players different abilities and efforts. Physical challenge often demands physical skills on speed, accuracy, endurance, dexterity and strength [12, 26]. Cognitive challenge requires mental abilities including memory, observation, reasoning, planning and problem solving issues [12, 26]. Emotional challenge, a relatively new one, requires players to deal with emotionally salient material or comprehend ambiguous elements by using cognitive effort rather than skill and dexterity [11]. Emotional challenge has recently been examined regarding many aspects including its difference from traditional challenges [4], the emotional responses it evokes [4, 24], its related game characteristics [4, 10] and related theories [10], etc. By resolving the tension within narrative, identifying characters, and exploring emotional ambiguities, players encountered with emotional challenge would be put in a more reflective state of mind and experience more diverse emotional experiences [4, 10, 11, 14, 24]. To explore the relationships between different challenges, an experimental study was conducted to investigate how emotional challenge shapes PX when separately from or jointly with conventional challenge [24]. Results show that adding emotional challenge to a game dominated by conventional challenge not only evokes more diverse emotional experiences, but also changes the emotional responses induced by the game dominated by conventional challenge.

Due to the efforts to better define game challenges, some subjective methods have been explored to evaluate challenge experience or its similar concept of demand experience. In both industry and research, challenge has long been assessed by interviewing playtesters in an unstructured way [14, 15]. Several challenge-related items are also contained in some widely used questionnaires for measuring broader PX, but the items are somehow fragmented, unspecific and

lack statistical validation [15]. In these early measures, challenge is considered more as a game characteristic rather than a player experience. However, it is not enough to measure challenge from the perspective of game designers as what players actually perceive is usually not the same with what a game tries to provide. Regarding this, a questionnaire of VGDS was developed to measure players' perceived physical, cognitive, emotional and social demand experiences in video games [5]. More recently, a questionnaire of CORGIS was developed and validated to directly measure different perceived challenges in video games. Through multiple rounds of interview and data analysis, challenge experience was finally operationalised into four kinds: cognitive, performative, emotional, and decision making [14]. Almost simultaneously, a video game challenge inventory was also built to identify players' distinct game challenge preferences and further a measurement instrument was developed for investigating the different challenge preferences [29].

Physiological sensing has long been leveraged to analyze or detect real-time PX, such as the state of flow [2, 3, 19] and multiple game-induced emotions [6, 20, 25]. Particularly, emotions such as boredom, flow and anxiety are often mapped with games of easy, appropriate and difficult level [8, 18]. With this mapping rule, various physiological signals, such as EEG, EDA, EMG, HRV, RSP and etc. are used to detect a specific emotion first and then determine whether the game is difficult or easy. This kind of method has widely been adopted to achieve dynamic difficulty adaptation in games dominated by traditional physical or cognitive challenges, such as Pong [13, 18], Tetris [7, 8] and first-person shooting games [21]. However, in modern digital games, emotional challenge often appears jointly with traditional kinds of challenges and results in complicated PX. This makes it difficult to detect difficulty or challenge just according to a certain kind of emotional experience. For example, if physiological signals indicate that a player is in a state of 'anxiety', it may be induced by a high level of difficulty, or it may also be caused by emotional challenge the player encountered. Therefore, challenge itself should be directly explored as a kind of player experience to understand how different challenges are perceived by players. With the recent efforts on subjective challenge measures which could provide metrics or labels of challenge experiences, we argue that it is a moment for exploring and developing an objective challenge measurement instrument based on physiological signals or other real-time game-induced data.

3 EXPERIMENT

To explore potential physiological signals to distinguish typical types of game challenges, we first choose three quests of Fallout 4¹ and constructed them as a game scenario to provides typical kinds of game challenges, then 12 participants were asked to play the game during which their physiological signals (EDA, ECG, EMG, RSP and TEM) were recorded. After playing the game, participants' self-reported challenge experiences, game immersion and multiple emotional responses were also collected. Physiological data were analyzed in the next step to detect different challenge types.

¹Detailed information can be found at https://fallout.fandom.com/wiki/Fallout_4

3.1 Game scenario

The first-person role-playing game of *Fallout 4* was chosen because it has narrative-rich contents, supports modification, and has proved to be a good candidate for studying emotional challenge [24]. With *Fallout 4*, three Brotherhood of Steel main quests including "Fire Support", "Call to Arms" and "Blind Betrayal"² were selected and then connected sequentially as our experimental scenario.

In the first quest of "Fire Support", the player meets Paladin Danse, a commander of the Brotherhood of Steel in a battle against a swarm of feral ghouls. In a conversation with Danse after the battle, the player knows that the Brotherhood is a just and united group and then responds Danse to assist the Brotherhood.

In the second quest of "Call to Arms", the player follows Danse to an abandoned rocket silo where he hopes to find a deep range transmitter to repair the Brotherhood of Steel communications array. Along the way, the player will fight alongside with Danse. In addition to encountering robbers and exploring the labyrinth-like silo, they will go through many fierce battles with the Brotherhood's biggest enemy—synthetic humans (i.e., synths). Throughout the quest, Danse will mention the crime of synths several times, which reflects Danse's extreme hatred of the synths. At the end of the quest, Danse will reward the player and the player responds to join the Brotherhood.

The third quest of "Blind Betrayal" takes place when the head of the Brotherhood, Elder Maxson, informs the player of the shocking news that Danse is a synth and ordered the player to execute Danse immediately. The player confirms the news with Proctor Quinlan and locates Danse with Scribe Haylen. When the player finds Danse hiding in a bunker, the player can have a conversation with Danse first and then decide to save (Event 19-a of Figure 1) or execute Danse (Event 19-b). Specifically in the conversation, Danse asks the player to execute him because he extremely despises synths. If the player decides to save Danse, he will convince Danse to escape with him. However, even if Danse is persuaded, they still have to face Maxson when they came out from the bunker. Again, the player must talk to Maxson to finally spare Danse (Event 20-a). Otherwise, Danse will be executed by Maxson (Event 20-b).

Figure 1 shows the timeline of the challenge-related events players need to get through in the scenario. These events were segmented by the authors according to the definition and characteristics of different types of challenges [4, 11, 14]. Particularly, each event was dominated by only a certain type of challenge. The emotional, physical and cognitive challenge types could be typically represented by the segmented events accordingly. The "Low-Meaningful Choices" could be a challenge type of Decision Making defined in CORGIS [14] but may not be so typical. The "Walk" was a control group which was characterized by very low level of challenge. Totally, 5 challenge types (PysC, CogC, EmoC, LMC and Walk) were included for further analyzing and detecting.

3.2 Participants and procedures

Regarding the difficulty for novices to play the game, twelve participants (11 male and 1 female, age $M=23.7$, $SD=2.32$) with 3D gameplay experiences but had not played *Fallout* series before were

recruited. Participants used a mouse and a keyboard to play the game on a PC with headphones (Figure 2-a). They were only allowed to use basic operations such as moving, shooting, loading ammunition and restoring health, as well as interacting with NPCs and key objects of the quests. Other advanced settings such as looting items, changing equipment and upgrading skills were not allowed.

Before each participant playing the game, a written consent was obtained, followed with written introductions of basic game operations and the scenario backgrounds. Then participants put on the physiological sensors and tried a test period to be familiar with the operations and settings. After that, participants rested for a while and started to play the game scenario. During the playing, their gameplay screening and physiological signals were recorded synchronously and an experimenter sat a little way off the participant to provide help when necessary. After playing, participants removed the physiological sensors and filled out multiple survey scales. Each of them was rewarded with a wireless headset. The whole experiment lasted about 100 minutes in which the playing of the scenario took about 50 minutes.

3.3 Data collection

Physiological signals including electrocardiography (ECG), electrodermal activity (EDA), respiratory activity (RSP), electromyography (EMG) and skin temperature (TEM) were recorded at a sampling rate of 1000 Hz using Biopac's MP150³. For ECG, EDA and EMG, disposable AgCl electrodes were used. Skin temperature was collected by using the TSD202B of Biopac and RSP data was acquired with a BioNomadix respiration transducer belt, also from Biopac. Figure 2-b shows the placements of the various electrodes for physiological recordings, and figure 2-c is an example of the recorded physiological signals.

Particularly, the CORGIS questionnaire [14] which consists 30 items (7-point Likert scale, 0 = strongly disagree and 6 = strongly agree) was adopted to evaluate types of cognitive, performative, emotional, and decision making challenge experiences. Note that the performative and physical challenge are identical with similar definition and characteristics. In addition, the 31-item Immersive Experience Questionnaire (IEQ, 7-point Likert scale, 0 = not at all and 6 = a lot) [17] was used to measure players' sense of immersion with respect to cognitive involvement, challenge, control, real-world dissociation and emotional involvement. To measure various emotional responses that might be induced, the Emotion Annotation and Representation Language (EARL) [27] was used to rate the level of 48 emotions (9-point Likert scale, 0 = not at all and 8 = quite a lot).

4 DATA ANALYSIS

The data of 2 participants, one with irregular heart rate and the other who did not play the game as required, were excluded. Data of the rest 10 participants were included for analyzing. Figure 3-a shows results of the 10 participants' self-report challenge experiences, immersive experiences, and major emotional responses. The average score of each kind of challenge was above point 3,

²Detailed walkthrough of the quests can be found at https://fallout.fandom.com/wiki/Fallout_4_quests

³Detailed information can be found at <https://www.biopac.com/>

Quest	Event	Dominated Challenge Type	Description of Event	Time Consuming (M, SD) /s
Fire	1	PysC	Battle against the ghouls.	59.3 (20.09)
Support	2	LMC	Have a conversation with Danse, knows about and responds to assist the Brotherhood.	134.1 (29.95)
Call to	3	Walk	Follow Danse to walk until encounter a gang of robbers on the way.	56.8 (8.05)
Arms	4	PysC	Fight against the robbers alongside with Danse.	13.9 (6.15)
	5	Walk	Follow Danse to walk until encounter a group of mutated mad dogs.	42.6 (24.35)
	6	PysC	Fight against the mad dogs alongside with Danse.	13.8 (7.66)
	7	LMC	Have a conversation with Danse to learn the crime of synths.	42.5 (9.40)
	8	CogC	Search for a password nearby to unlock a door in the way.	130.5 (51.87)
	9	PysC	Fight alongside with Danse against a group of synths coming from the opening door.	17.9 (3.54)
	10	PysC	Fight alongside with Danse against a group of synths coming from another room.	36.5 (11.00)
	11	PysC	Fight alongside with Danse against the fire power of synths' weapon system.	32.4 (7.63)
	12	CogC	Find the equipment to restore the power supply to the silo.	30.8 (16.43)
	13	CogC	Crack the password on the terminal to restore the power to the silo.	83.0 (38.39)
	14	PysC	Fight alongside with Danse against a large number of synths from the bottom of the silo.	31.7 (12.18)
	15	LMC	Have a conversation with Danse and responding to join the Brotherhood.	92.0 (17.54)
Blind	16	EmoC	Have a conversation with Maxson, be told Danse is a synth and be ordered to execute Danse	125.2 (7.05)
Betrayal	17	EmoC	Have a conversation with Proctor Quinlan and be confirmed that Danse is a synth.	141.5 (31.12)
	18	EmoC	Have a conversation with Scribe Haylen and learn where Danse is hiding.	102.6 (23.01)
	19-a	EmoC	Have a conversation with Danse until convincing Danse to escape.	197.5 (33.54)
	19-b	EmoC	Have a conversation with Danse until executing Danse.	138.0 (19.80)
	20-a	EmoC	Have a conversation with Maxson outside until convincing Maxson to spare Danse.	243.2 (41.88)
	20-b	EmoC	Witness Maxson executing Danse.	13.5 (2.12)

*PhsC = Physical Challenge; *CogC = Cognitive Challenge; *EmoC = Emotional Challenge; *LMC = Low-Meaningful Choices

Figure 1: Timeline of the challenge-related events.

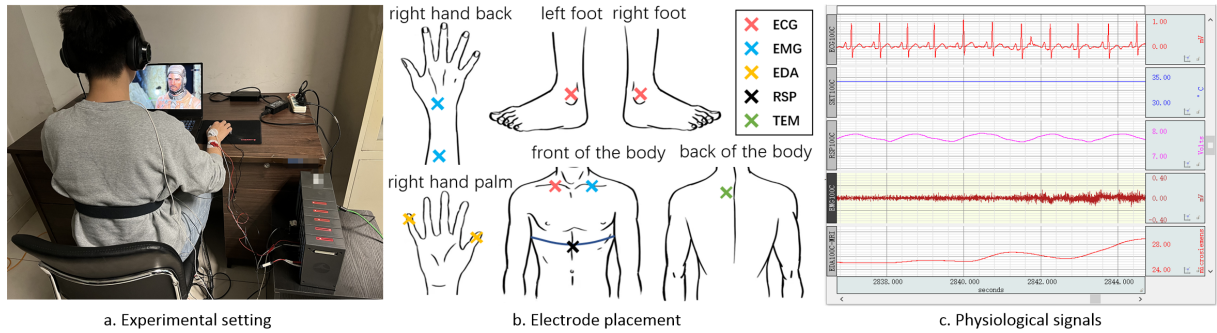


Figure 2: Experimental setting, electrode placement, and physiological signals.

indicating that the game scenario has successfully induced participants' four kinds of challenge experiences, which were also kept at a similar level. Results also show that participants immersed in the game and experienced a variety of emotions including trust, love, hope, doubt, courage, anxiety, etc.

Different participants spend different time playing the game scenario. We watched each participant's gameplay screening and marked the start and end time of each challenge-related event of Figure 1, from which we got the consuming time of each event and also located each participant's physiological signals of each event.

4.1 Physiological signal processing

Signal pre-processing was conducted with AcqKnowledge software of Biopack⁴. Specifically, for ECG data, z-score normalization was first applied to reduce individual differences. Median filters of 200-ms and 600-ms width were used to remove baseline wander caused by the participant's movement or breathing [16]. Then, R-wave peaks were detected by Pan-Tompkins QRS detector [23], and the

RR intervals (heart rate variability, HRV) were computed as the time between two successive R-wave peaks. For EDA data, baseline wander was removed by a high pass filter [28] with 0.02 Hz. For RSP data, a band-pass filter with 0.1 Hz to 5 Hz was used to remove unusual breathing cycles. Penh Analysis of Acqknowledge was then used to detect respiratory circulation by setting the 65% of exhalation volume as threshold.

Signal features were then extracted using Matlab. For each kind of signal of each participant, z-score normalization was applied first to reduce individual differences. Then, each frame of physiological features were computed using signals inside a sliding window with 10-second width and 5-second shift. A total of 80 features were extracted and shown by figure 3-b. Specifically, ECG time-domain features were computed using the R-wave peak values, RR intervals, as well as their first-order differences (Diffs) between two successive tokens. To compute ECG frequency-domain features, the RR intervals were re-sampled to 8 Hz using a cubic spline interpolation [22]. Then, Fast Fourier Transform-based method on 256 samples each frame (reset to 32-second window width) were applied and

⁴Detailed information can be found at <https://www.biopac.com/product/acqknowledge-software/>

CORGIS		M (SD)
Cognitive challenge		3.95 (0.68)
Emotional challenge		4.44 (0.55)
Performative challenge		4.06 (0.74)
Decision making		4.38 (0.57)
IEQ		M (SD)
Cognitive involvement		4.09 (0.74)
Real-world dissociation		3.90 (0.72)
Control		3.38 (0.61)
Emotional involvement		3.95 (1.11)
Challenge		3.40 (0.83)
Emotional Responses		
Top 1 & 2	Trust & Love	
Top 3 & 4	Hope & Doubt	
Top 5 & 6	Courage & Anxiety	
Top 7 & 8	Serenity & Empathy	
Top 9 & 10	Tension & Delight	

		HR	BPM (Bit Per Minute)
ECG	Time	R	Mean, Std, MAD
		Diff R	Mean, Std, MAD, MAV
		RR (HRV-)	Mean, Std, MAD
		Diff RR	Mean, Std, MAD, MAV
	Frequency (HRV-)	RR power	VLF, LF, HF, TP, LF/HF, LF norm, HF norm
		RR PSD	VLF, LF, HF, TP, LF/HF, LF norm, HF norm
TEM	Amplitude		Mean, Std, MAD
		Diff Amplitude	Mean, Std, MAD, MAV
RSP	Time	Amplitude	Mean, Std, MAD
		Diff	Mean, Std, MAD, MAV
		Period	Mean, Std, MAD, BPM (Breath Per Minute)
		Diff Period	Mean, Std, MAD, MAV
EMG	Wavelet decomposition		1-7 Band power, Total power, 1-7 Band power%, Entropy, RMS
EDA	Time		LF SAV, HF SAV, VHF SAV
		Frequency	LF SAV, LF power, LF std power, HF SAV, HF power, HF power std, VHF SAV, VHF power, VHF std power

a. Subjective ratings

b. Physiological features

Figure 3: Participants' self-report experiences and the extracted physiological features. Abbreviations: Std = standard deviation, MAD = mean absolute deviation, MAV = mean absolute value, TP = total power or PSD of {VLF, LF, HF}, LF or HF norm = $\frac{\text{LF or HF}}{\text{total power or PSD of \{LF, HF\}}}$, Band power% = $\frac{\text{Band power}}{\text{Total power}}$, Entropy = $-\sum \text{Band power\%} \log \text{Band power\%}$, RMS = root mean square, SAV = sum absolute value.

the frequency bands were grouped into three ranges: very-low-frequency (VLF) (0–0.04 Hz), low-frequency (LF) (0.04–0.15 Hz), and high-frequency (HF) (0.15–0.4 Hz) [16]. The power and power spectral density (PSD) in these ranges were utilized as features. For RSP data, the same time-domain features as ECG data were extracted. For TEM data, the amplitudes and their Diffs were made use. For EMG data, the signal was first down-sampled to 32 Hz, followed by a 6-level wavelet decomposition with Daubechies5 [9]. For EDA data, time- and frequency-domain features [28] of low-frequency (LF) (0.02–0.5 Hz), high-frequency (HF) (0.5–1 Hz), and very-high-frequency (VHF) (> 1 Hz) ranges were computed.

Finally, the feature frames located in the event periods (marked by start and end time) of five challenge types (PysC, CogC, EmoC, LMC and Walk) were collected and deployed for analysis. Frame location was indicated by start time of the window. Totally 2941 frames over the 10 participants were used and normalized again by z-score for each participant.

4.2 Feature selection and detection results

To investigate the ability of different physiological features for representing challenge information, we adopted two alternative methods of feature selection: ANOVA-based and Regression-based. In ANOVA-based feature selection, one-way ANOVA analysis was first applied to each feature for each participant (80-feature $\times$ 10-participant). Only the features resulting in significant difference ($p < 0.05$) among challenge types were kept. Second, the remaining features were used to compute the Pearson correlation within each participant. Only one of each feature pair was kept when they were almost identical and produced a correlation > 0.95 or < -0.95 . Finally, a feature was selected if it was kept for at least one participant. 58 features out of 80 were selected and shown by bold font in figure 3-b.

In Regression-based feature selection, we applied the participant-independent logistic regression to each challenge by treating its

label as 1 and others as 0. Namely, five logistic regression models (with respect to the five challenge types) were fitted over the 80-dimensional features of 10 participants. The model coefficient value represented the increment of the chance of being predicted to the target challenge, evaluated by $\text{logit} = \log \frac{\text{probability}}{1-\text{probability}}$, when the corresponding feature was increased by 1. In our analysis, to eliminate the possible impact of unbalanced data, we forced the interception coefficients to be zero so that the models focused more on feature effects instead of data bias. Accordingly, 29 features were selected by keeping only the features with significant coefficients ($p < 0.05$) for at least one challenge, where 25 features were overlapped with the 58 features selected by ANOVA-based method. The selected features were shown by orange font in figure 3-b, and their coefficients were shown in figure 4. The +/- coefficient indicated a significantly +/- correlation between the feature and the chance of being predicted to the challenge type.

Finally, we evaluated regression performance of the two feature selection methods by 10-fold cross validation. The data of 10 participants were first divided into 10 batches randomly. The logistic regression models introduced above were fitted and evaluated 10 times using the selected features. In each time, the five models were trained on 9 batches of data and evaluated on the remaining 1 batch (as test data). During evaluation, each frame in test data was predicted to the challenge type (model) with the highest logit. The results were summarized in figure 5-a. When using ANOVA-based feature selection, an average accuracy of 69.3% was achieved. The confusion matrix is shown in figure 5-b, which demonstrated the prediction accuracy around 80% on emotional and physical challenge. The Regression-based selection obtained an average accuracy of 68.6%. These results significantly outperformed random choice over the five classes.

Considering that the 10-fold cross validation did not reflect individual differences, we also evaluated the regression performance by leave-one-subject-out cross validation: in each time, data of one participant was deployed as test data while using others as

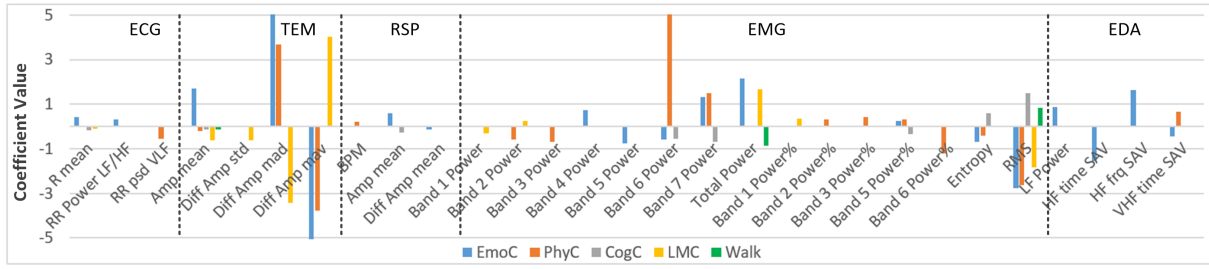


Figure 4: Coefficients selected by logistic regression models. The amplitudes shown in the figure were cut-off at ± 5 . The $\pm$ -value of coefficient indicated a significantly $\pm$ -correlation between the feature and the chance of being predicted to the challenge type.

training data. The average accuracy were reported in figure 5-a. Performance degradation was observed compared to the 10-fold validation. Moreover, Regression-based feature selection yielded the accuracy of each participant ranging from 34.6% to 77.5%. These results suggests that mismatch exists among different participants.

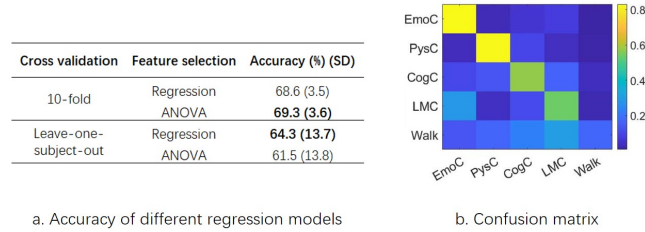


Figure 5: Cross validation results of logistic regression models.

5 DISCUSSION

This study conducted a primary experiment to probe into the potential of using physiological signals (EDA, ECG, EMG, RSP and TEM) to distinguish different types of game challenges. Results show that more than half of the extracted features could contribute to the detection of challenge types. It seems that TEM-related features contribute most to detect emotional challenge and EMG-related features contribute most for detecting physical challenge. It is inspiring that the prediction accuracy of emotional and physical challenge reaches to 80%, which suggest potential for further development of a real-time challenge measurement instrument based on physiological signals.

Meanwhile, other challenge types have not achieved such high accuracy, which may be because the extracted physiological features could not indicate them clearly or the feature frames are not sufficient enough with the single game scenario used. As each type of challenge may be represented by various forms of game events, more genres of games and scenarios should be included to identify specific physiological information of a certain type of challenge. Although a game event may dominated by only one type of challenge and thus be expected to induce a major type of challenge experience, players may also have other kind of challenge experiences, especially when different challenges are correlated with

each other in the game. Moreover, different players with different abilities and motivation may also perceive the same game challenge setting at different levels. Therefore, to develop a real-time challenge measurement instrument, it is important to get more fine-grained individual challenge experience, regarding both challenge types and levels as training metrics, and also address the above-mentioned limitations by including more game scenarios, using balanced challenge data, trying more physiological features and exploring advanced AI models, etc.

ACKNOWLEDGMENTS

This work was supported by the National Key R&D Program of China (Grant No. 2020YFC2004100).

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